

MICHAEL JAMES LAMBERT

Personal Site: <https://michaeljlambert.github.io/>

8 First Ave, #2A, Montpelier, VT, 05602

mlamber2@norwich.edu, michael.lambert@umb.edu, michael.james.lambert@gmail.com

EMPLOYMENT

Norwich University Lecturer of Mathematics	<i>Aug. 2024-ongoing</i>
University of Massachusetts-Boston Lecturer of Mathematics	<i>Sept. 2020-ongoing</i>
Champlain College Adjunct Instructor	<i>Aug. 2023-May 2024</i>
Topos Institute Research Associate	<i>June-Aug. 2022</i>
Mount Allison University Postdoctoral Research Associate	<i>Sept. 2021-April 2022</i>
Dalhousie University Graduate Student	<i>Sept. 2014-Oct. 2019</i>
University of Missouri-Columbia Graduate Student Instructor	<i>Aug. 2011-May 2014</i>

EDUCATION

Dalhousie University PhD Mathematics Thesis: “An Elementary Account of Flat 2-Functors”	<i>2014-2019</i>
University of Missouri-Columbia MA Mathematics	<i>2011-2014</i>
University of Vermont MS Mathematics	<i>2009-2011</i>
Dartmouth College BA English Literature, Philosophy	<i>2004-2007</i>

PAPERS AND PREPRINTS

- Twisted double functors and loosely discrete opfibrations, with Evan Patterson and David Jaz Myers. ArXiv preprint, 2026.
- A Topos-Theoretic Semantics of Intuitionistic Modal Logic with an Application to the Logic of Branching Spacetime, Accepted in *Applied Categorical Structures*, 2024.
- Representing Knowledge and Querying Data using Double-Functorial Semantics, with Evan Patterson. Proceedings Seventh International Conference on Applied Category Theory 2024 (ACT 2024), Oxford, United Kingdom, 17-21 June 2024, Electronic Proceedings in Theoretical Computer Science 429, pp. 174–189, Sept. 2025.
- Discrete 2-Fibrations
Higher Structures, 8(1), 54-96, May 2024.

- Cartesian double theories: A double-categorical framework for categorical doctrines, with Evan Patterson. *Advances in Mathematics*, 444, May 2024.
 - Double Fibrations, with Geoff Cruttwell, Dorette Pronk, and Martin Szyld. *Theory and Applications of Categories*, 38(35), 2022.
 - Double Categories of Relations
Theory and Applications of Categories, 38(33), 2022.
 - Discrete Double Fibrations
Theory and Applications of Categories, 37(22), 2021.
- Other Preprints:*
- A Topos View of Blockchain Consensus Protocols
2021 preprint: arxiv.org/abs/2111.07461
 - Representation Embeddings of Cartesian Theories
2019 preprint: arxiv.org/abs/1612.02497
 - A 2-Site for Continuous 2-Group Actions
2019 preprint: arxiv.org/abs/1911.05141
 - Computing Weighted Pseudo-Colimits
2019 preprint: arxiv.org/abs/1711.05903
 - An Elementary Account of Flat 2-Functors
2019 Dalhousie University, PhD Thesis
DalSpace Link: dalspace.library.dal.ca/handle/10222/75671
 - Undecidability of the Theory of $k\langle X, Y \rangle$ -Modules
2014 University of Missouri-Columbia, MA Thesis

TALKS

Data Queries via Double Categorical Functorial Semantics

FMCS 2026, June 2026.

Twisted Double Functors

CMS Summer Meeting, June 2026.

An Introduction to Category Theory with an Eye Toward Applications in Data Science

Norwich University Mathematics Colloquium, 2 Oct, 2024.

Representing Knowledge and Querying Data using Double-Functorial Semantics

ACT 2024, 17 June, 2024.

A cartesian equipment bestiary

FMCS, 9-10 June 2023.

Double Categories of Relations

Virtual Double Categories Workshop, 2 December 2022.

Cartesian Equipments as Databases

AtCat Seminar, Dalhousie University, 18 October 2022

Regular Logic in a ‘Double Category of Relations’

Peripatetic Seminar, University of Calgary, 17 March, 2022.

Improvements to ‘Double Categories of Relations’

AtCat Seminar, Dalhousie University, 8 February 2022

Blockchain Safety in a Topos

Ottawa Logic Seminar, University of Ottawa, 2 December, 2021

A Topos View of Blockchain Safety

AtCat Seminar, Dalhousie University, 16 November, 2021

Discrete Double Fibrations

CT20→21, University of Genoa, 30 August, 2021

Characterizing Double Categories of Relations

Applied Category Theory 2021, Cambridge University, 16 July 2021

Discrete Double Fibrations

Peripatetic Seminar, University of Calgary, 12 February 2021

A 2-Site for Continuous 2-Group Actions

CMS Summer Meeting, University of Regina, 10 June 2019

A Site for Continuous Group Actions

AtCat Seminar, Dalhousie University, 5 March 2019

Exactness of an Internal Colimit Functor for 2-Categories

AtCat Seminar, Dalhousie University, 20 November 2018

The Calculus of Fractions in a Cosmos of Internal Categories

Octoberfest 2018, City College CUNY, 28 October 2018

Diaconescu's Theorem for 2-Toposes of Stacks

Category Theory 2018, University of Azores, 13 July 2018

A Tensor Product of Fibrations as a Codescent Object

CMS Summer Meeting 2018, University of New Brunswick, 4 June 2018

Flat Category-Valued Pseudo-Functors

AtCat Seminar, Dalhousie University, 30 January 2018

Principal Bundles and Torsors as Flat Functors

AtCat Seminar, Dalhousie University, 28 November 2017

Pseudo-Functors, Generalized Principal Bundles, and Torsors

Octoberfest 2017, Carnegie Mellon University, 28 October 2017

Generalizing Principal Bundles

Category Theory 2017, University of British Columbia, 18 July 2017

Wild Algebras in Cartesian Categorical Logic

TACL 2017, Prague, Czech Republic, 30 June 2017

Representation Embeddings of Cartesian Theories

AtCat Seminar, Dalhousie University, 10 Jan 2017

Representation Embeddings of Algebraic Theories

Union College Mathematics Conference, Union College, 3 December 2016

TEACHING

As Instructor

MA122: Calculus II, Norwich University

MA212: Finite Mathematics, Norwich University

MA095: Intermediate Algebra, Norwich University

MA107: Precalculus, Norwich University

Math 125: Precalculus, Champlain College
Math 230: Calculus I, Champlain College
Math 240: Calculus II, Champlain College
Math 115: College Algebra, University of Massachusetts-Boston
Math 125: Introductory Statistics, University of Massachusetts-Boston
Math 129: Precalculus for Management, University of Massachusetts-Boston
Math 130: Precalculus, University of Massachusetts-Boston
Math 134: Calculus for Management, University of Massachusetts-Boston
Math 140: Calculus I, University of Massachusetts-Boston
Math 141: Calculus II, University of Massachusetts-Boston
Math 2002: Intermediate Calculus II, Dalhousie University
Math 1100: College Algebra, University of Missouri-Columbia

As Teaching Assistant

Math 1000: Calculus I, Dalhousie University
Math 1010: Calculus II, Dalhousie University
Math 1290: Engineering Mathematics II, Dalhousie University,
Math 1280: Engineering Mathematics I, Dalhousie University
Math 1500: Calculus I, University of Missouri-Columbia

AWARDS

Nova Scotia Graduate Scholarship \$30,000 CAD, Dalhousie University	2014-2017
AARMS Student Travel Award Homotopy Type Theory Workshop The Fields Institute, University of Toronto \$1,000 CAD	June 2016
AARMS Student Travel Award AARMS Summer School 2014, Dalhousie University, \$650 CAD	July 2014
Excellence in Teaching Award 2013-2014 Mathematics Department University of Missouri-Columbia	April 2014